Electronic Devices and Systems

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

setu.ie

Electronic Devices and Systems (A27463)

Short Title:	Electronic Devices and Systems		
Faculty	Science and Computing		
Department	Science		
Cluster	Mathematics and Physics		
Author	CKEARY		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces the student to digital devices and systems. The structure and principle of operation of bipolar and field effect devices will be described. The fundamental building blocks of digital design across several levels of abstraction, from semiconductor devices to logic gates to registers, will be introduced. The student will learn how to design a range of basic combinational and sequential circuits with implementation at logic circuit level. This will be accompanied by a comprehensive study of digital electronics and digital circuits in the laboratory setting.

Programmes

`ELEC-0021`	BSc (Hons) in Applied Computing (`WD_KCOMP_B`)	2 / 3 / E
`ELEC-0021`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	2 / 3 / M
`ELEC-0021`	BSc (Hons) in the Internet of Things (`WD_KINTT_B`)	2 / 3 / M

Indicative Content

- Bipolar devices: review of pn junction diode and characteristics; bipolar junction transistor: structure, principle of operation and characteristics
- Field effect devices: field effect transistor (MOSFET and CMOS); structure, principle of operation and characteristics
- Digital devices: digital concepts and components, logic families
- Combinational logic: circuits and design
- Sequential logic: latches and flip-flops, registers, asynchronous and synchronous counters; waveform/timing diagrams
- Decoders, demultiplexers and multiplexers

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe and explain the structure and principle of operation of bipolar and field effect devices.
2. Describe a range of digital devices based on combinational and sequential logic.
3. Discuss a range of design issues in digital devices; plot and analyze timing diagrams.
4. Demonstrate problem solving and investigative skills in conjunction with the integrated practical programme, and present results in the form of a report.

Learning and Teaching Methods

- Lectures: The lectures will be used to present new topics and their related concepts. Techniques used to design a range of combinational and sequential logic circuits and devices will be introduced in lectures.
- Laboratory-based practicals: An integrated practical programme is designed to run in parallel with lectures. This will allow the student to develop experimental skills in the design and construction of a

range of combinational and sequential circuits. The importance of planning of experiments, data acquisition, and data analysis will be emphasised throughout.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3
Continuous Assessment	30%	4
Practical	30%	4

Assessment Criteria

<40%: Unable to demonstrate a basic understanding of elementary digital devices and systems. An understanding of very little of the course content.

40%-49%: Able to produce diagrams of digital devices and systems but unable to explain clearly principle of operation. Able to provide partial solutions to circuit problems.

50%-59%: Able to describe clearly principles of operation of a range of digital circuits. Able to analyse digital devices and systems and provide solutions to circuit problems. Demonstrates an understanding of the mathematical techniques used in circuit analysis.

60%-69%: Able to apply problem-solving techniques to new similar circuit problems. Demonstrates an ability to design a digital system from a number of component parts.

70%-100%: All the previous to an excellent level. Demonstrates an understanding of the limitations of digital circuits and systems and subsequent consequences.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	24	
Independent Learning	75	

Supplementary Material

- Floyd, T. _Digital Fundamentals_. 11th ed. USA: Prentice-Hall, 2014.
- Harris, D. and S. Harris. _Digital Design and Computer Architecture_. 2nd ed. USA: Morgan Kaufmann, 2012.
- Storey, N. _Electronics: A Systems Approach_. 5th ed. UK: Pearson, 2013.
- Wakerly, J. _Digital Design: Principles and Practices_. 4th ed. USA: Prentice-Hall, 2005.

Requested Resources

- SCIENCE LAB: Physics